



NARAYANA
IIT-JEE ACADEMY-INDIA

JR IIT CO SUPER CHAINA

Date: 29-06-2024

Time: 3Hrs.

WAT-3

Max. Marks: 198

JEE-ADVANCE-2020-P1-Model

IMPORTANT INSTRUCTIONS

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec- I(Q.N: 1 – 6)	Questions with Single Correct Options	+3	-1	6	18
Sec- II(Q.N: 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec- II(Q.N: 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				18	66

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec- I(Q.N: 19 – 24)	Questions with Single Correct Options	+3	-1	6	18
Sec- II(Q.N: 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec- II(Q.N: 31– 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				18	66

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec- I(Q.N: 37 – 42)	Questions with Single Correct Options	+3	-1	6	18
Sec- II(Q.N: 43 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-1	6	24
Sec- II(Q.N: 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				18	66

PAPER SETTER :: KNL – N-120

PHYSICS

MAX.MARKS: 66

SECTION – 1 (Maximum Marks: 18)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer. For each question, choose the correct option corresponding to the correct answer.

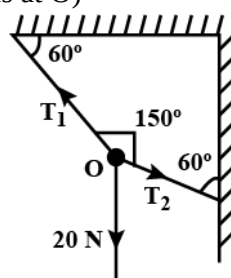
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks : 0 If none of the option is chosen.(i.e the question is un answered)

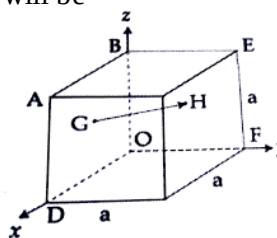
Negative Marks : -1 In all other cases.

- The resultant of two vectors $\vec{P}$ and $\vec{Q}$ is $\vec{R}$. If the magnitude of $\vec{Q}$ is doubled the new resultant becomes perpendicular to $\vec{P}$. Then magnitude of $\vec{R}$ is
 1) $\frac{P^2 - Q^2}{2PQ}$ 2) $\frac{P+Q}{P-Q}$ 3) Q 4) $\frac{P}{Q}$
- Two vectors $\vec{a}$ and $\vec{b}$ have equal magnitudes of 12 units. These vectors are making angles 30° and 120° with the x-axis respectively. Their sum is $\vec{r}$. Then the x and y components of $\vec{r}$ are
 1) $(6\sqrt{3}-6), (6+6\sqrt{3})$ 2) $(6\sqrt{3}+6), (6+6\sqrt{3})$
 3) $(6\sqrt{3}-6), (6-6\sqrt{3})$ 4) $(6\sqrt{3}+6), (6-6\sqrt{3})$
- Cosines of angles made by a vector with X, Y are $3/5\sqrt{2}, 4/5\sqrt{2}$ respectively. If the magnitude of the vector is $10\sqrt{2}$, then the vector is
 1) $8\hat{i} + 6\hat{j} - 10\hat{k}$ 2) $6\hat{i} - 8\hat{j} - 10\hat{k}$ 3) $-6\hat{i} - 8\hat{j} + 10\hat{k}$ 4) $6\hat{i} + 8\hat{j} + 10\hat{k}$
- If $\vec{A} = 2\hat{i} + 3\hat{j} + 6\hat{k}$ and $\vec{B} = 3\hat{i} - 6\hat{j} + 2\hat{k}$ then vector perpendicular to both $\vec{A}$ and $\vec{B}$ has magnitude K times that of $6\hat{i} + 2\hat{j} - 3\hat{k}$. Then K is
 1) 1 2) 3 3) 7 4) 9
- If 'O' is at equilibrium, then the values of the tensions T_1 and T_2 are (20 N is acting vertically downwards at O)



- 1) 20N, 30N 2) $20\sqrt{3}N, 20N$ 3) $20\sqrt{3}N, 20\sqrt{3}N$ 4) 10N, 30N
- In a cube of side length 'a' shown in the figure, the vector from the central point of the face ABOD to the central point of the face BEFO will be

- 1) $\frac{1}{2}a(\hat{i} - \hat{k})$ 2) $\frac{1}{2}a(\hat{j} - \hat{i})$
 3) $\frac{1}{2}a(\hat{k} - \hat{i})$ 4) $\frac{1}{2}a(\hat{j} - \hat{k})$



SECTION – 2 (Maximum Marks : 24)

This section contains SIX (06) questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

7. The magnitude of the vector product of two vectors $\vec{A}$ and $\vec{B}$ may be :
 1) greater than AB 2) equal to AB 3) less than AB 4) equal to zero
8. The following sets of three vectors act on a body, whose resultant can be zero. These are :
 1) 10, 10, 10 2) 10, 10, 20 3) 10, 20, 20 4) 10, 20, 40
9. For two vectors $\vec{A}$ and $\vec{B}$, identify the correct relation :
 1) $\vec{A} + \vec{B} = \vec{B} + \vec{A}$ 2) $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ 3) $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$ 4) $(\vec{A} - \vec{B}) \times (\vec{B} - \vec{A}) = 0$
10. Regarding vectors, which of the following is a correct statement?
 1) Two equal vectors can never give zero resultant
 2) Three non-coplanar vectors can not give zero resultant
 3) If $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$ and $|\vec{a}| \neq |\vec{b}| \neq |\vec{c}|$, then $\vec{a} + \vec{b} + \vec{c}$ can never be a null vector
 4) If $\vec{a} \times \vec{b} = 0$ and $|\vec{a}| = |\vec{b}|$, then $\vec{a} + \vec{b}$ can be zero
11. Which of the following vectors is/are perpendicular to the vector $5\hat{k}$?
 1) $4\hat{i} + 3\hat{j}$ 2) $6\hat{i}$ 3) $7\hat{k}$ 4) $3\hat{i} + 4\hat{j}$
12. Let three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are related as $\vec{a} + \vec{b} = \vec{c}$. The angle between $\vec{a}$ and $\vec{b}$ is θ .
 Select the correct alternative :
 1) Angle between $\vec{c}$ and $\vec{b}$ is greater than $\theta/2$ if $|\vec{a}| > |\vec{b}|$
 2) Angle between $\vec{c}$ and $\vec{a}$ is greater than $\theta/2$ if $|\vec{a}| > |\vec{b}|$
 3) Angle between $\vec{c}$ and $\vec{b}$ can be $\pi - \theta$, for some values of $\vec{a}, \vec{b}$
 4) Angle between $\vec{c}$ and $\vec{b}$ cannot be $\pi - \theta$, for some values of $\vec{a}, \vec{b}$

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If ONLY the correct numerical value is entered as answer.

Zero Marks : 0 In all other cases.

13. Three vectors $\vec{F}_1 = (3\hat{i} + 2\hat{j} - \hat{k})N$, $\vec{F}_2 = (3\hat{i} + 4\hat{j} - 5\hat{k})N$ and $\vec{F}_3 = A(\hat{i} + \hat{j} - \hat{k})N$ act simultaneously on a particle. In order that the particle remains in equilibrium, the value of A should be $-x$ then find x value is ____
14. A force of 10N is resolved into the perpendicular components. If the first component makes 30° with the force, the magnitudes of the components are $x\sqrt{3}(N)$ and $y(N)$ respectively. Then $x + y$ is (N)
15. What is the displacement of the point of a wheel initially in contact with the ground when the wheel rolls forward half a revolution? Take the radius of the wheel as R and the x-axis as the forward direction is $\tan^{-1}\left(\frac{x}{\pi}\right)$ with x-axis then x is ____
16. A particle is moving eastwards with a velocity of 5 m/s. In 10s the velocity changes to 5 m/s northwards. Find the average acceleration in this time is $\frac{1}{\sqrt{x}}$ m/sec² then x is ____
17. Two vectors $\vec{A}$ and $\vec{B}$ are defined as $\vec{A} = a\hat{i}$ and $\vec{B} = a(\cos \omega t \hat{i} + \sin \omega t \hat{j})$, where a is a constant and $\omega = \pi / 6 \text{ rad s}^{-1}$. If $|\vec{A} + \vec{B}| = \sqrt{3}|\vec{A} - \vec{B}|$ at time $t = \tau$ for the first time, the value of τ , in seconds, is ____.
18. The vectors $\hat{i} + 2\hat{j} + 3\hat{k}$, $-3\hat{j} + 2\hat{k}$ and $-5\hat{k}$ represent the edges of a parallelepiped; obtain its volume is x units then x is ____

CHEMISTRY**MAX.MARKS: 66****SECTION – 1 (Maximum Marks: 18)**

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Negative Marks : -1 In all other cases.

19. Ratio of frequency of revolution of electron in the 2nd excited state of He⁺ and 2nd state of hydrogen is :
- 1) $\frac{32}{27}$ 2) $\frac{27}{32}$ 3) $\frac{1}{54}$ 4) $\frac{27}{2}$
20. A photon of wavelength 4×10^{-7} m strikes on metal surface. The work function of the metal is 2.13 eV. The velocity of the photo electron is
- 1) $5.67 \times 10^6 \text{ ms}^{-1}$ 2) $5.67 \times 10^5 \text{ ms}^{-1}$ 3) $5.67 \times 10^{-5} \text{ ms}^{-1}$ 4) $5.67 \times 10^{-6} \text{ ms}^{-1}$
21. Which of the following postulates does not belong to Bohr's model of the atom?
- 1) Angular momentum of electron is an integral multiple of $\frac{h}{2\pi}$
- 2) The electron stationed in the orbit is stable
- 3) The path of an electron is circular
- 4) Mass of electron increases with increase velocity
22. The shortest wavelength of H atom in the Lyman series is I. The longest wavelength in the Balmer series of He⁺ is :
- 1) $\frac{9\lambda}{5}$ 2) $\frac{36\lambda}{5}$ 3) $\frac{5\lambda}{9}$ 4) $\frac{5}{9\lambda}$
23. The radius of the second Bohr orbit, in terms of the Bohr radius, a_0 , in Li²⁺ is :
- 1) $\frac{4a_0}{3}$ 2) $\frac{4a_0}{9}$ 3) $\frac{2a_0}{3}$ 4) $\frac{2a_0}{9}$
24. The number of photons of light having wave number 'x' in 10 J of energy source is
- 1) $10hc x$ 2) $\frac{hc}{10x}$ 3) $\frac{10}{hc x}$ 4) none of these

SECTION – 2 (Maximum Marks : 24)

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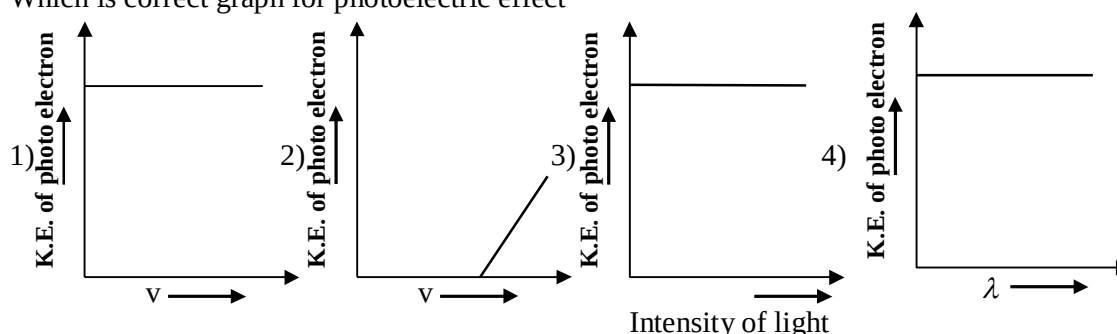
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25. Which is correct graph for photoelectric effect



26. The energy of an electron in the first Bohr orbit of H atom is -13.6 eV. The impossible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is/are

- 1) -3.4 eV 2) 4.2 eV 3) -6.8 eV 4) $+6.8$ eV

27. The number of correct statement/s about the black body from the following is ____

- 1) Emit or absorb energy in the form of electromagnetic radiation.
- 2) Frequency distribution of the emitted radiation depends on temperature
- 3) At a given temperature, intensity VS frequency curve passes through a maximum value
- 4) The maximum of the intensity VS frequency curve is at higher frequency compared to the at lower temperature

28. According to Bohr's atomic theory, select the correct statements

- 1) Kinetic energy of electron is $\propto \frac{Z^2}{n^2}$
- 2) The product of velocity (v) of electron and principal quantum number (n), ' vn ' $\propto Z^2$.
- 3) Frequency of revolution of electron in an orbit is $\propto \frac{Z^3}{n^3}$
- 4) Coulombic force of attraction on the electron $\propto \frac{Z^3}{n^4}$

29. In a sample of H-atoms electrons are de-excited from 4th excited state to ground state. Which is/are correct statement?
- 1) No line observed in p-fund series.
 - 2) Total ten lines observed in spectrum
 - 3) 4 line in UV-region and 3 line in visible region observed
 - 4) One line observed in Brackett series
30. The angular momentum of electron can have the value(s) ?
- 1) $0.5 \frac{h}{\pi}$
 - 2) $\frac{h}{\pi}$
 - 3) $\frac{h}{0.5\pi}$
 - 4) $2.5 \frac{h}{2\pi}$

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-
31. The value of 'n' of the highest excited state that an electron of hydrogen atom in the ground state can reach when 12.09 eV energy is given to the hydrogen atom is
32. If the radiation source has the duration of 2 nano seconds and the number of photons emitted is 2.5×10^{15} . The energy of the source is $x \times 10^{-10} J$. The value of x is (Nearest integer)
33. In an collection of H-atoms, all the electrons jump from $n = 5$ to ground level finally (directly or indirectly), without emitting any line in Balmer series. The number of possible different radiations is :
34. The energy of one mole of photons of radiation of wavelength 300 nm is ____ kJ/mole
35. The ratio of velocity of the electron in the third and fifth orbit of Li^{2+} would be x then 3x is ____
36. The number of spectral lines produced when an electron jumps from 5th orbit to 2nd orbit in the hydrogen atom is ____

MATHS**MAX.MARKS: 66****SECTION – 1 (Maximum Marks: 18)**

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37. In a triangle ABC, angle B is right angled, $AC = 2$ and $A(2, 2)$ $B(1, 3)$ then the length of the median AD is

- 1) $\frac{1}{2}$ 2) $\sqrt{\frac{5}{2}}$ 3) $\frac{5}{\sqrt{2}}$ 4) $\frac{1}{\sqrt{2}}$

38. If $K\sqrt{-1}$ is a root of the equation $x^4 + 6x^3 - 16x^2 + 24x - 80 = 0$ then $K = \underline{\hspace{1cm}}$

- 1) ± 1 2) ± 2 3) ± 3 4) ± 4

39. Value of $\sin^6 7\frac{1}{2} + \cos^6 7\frac{1}{2} = \frac{10+3\sqrt{3}}{K}$ then $K = \underline{\hspace{1cm}}$

- 1) 4 2) 7 3) 16 4) NT

40. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a-c) + ad(a-d) + bc(b-c) + bd(b-d)$ is

- 1) 0 2) 8000 3) 8080 4) 16000

41. If $\cot \alpha = 1$ and $\sec \beta = \frac{-5}{3}$ whose $\pi < \alpha < \frac{3\pi}{2}$ and $\frac{\pi}{2} < \beta < \pi$. Then the value of $\tan(\alpha + \beta)$ and the quadrant in which $\alpha + \beta$ lies respectively are

- 1) $-\frac{1}{7}$ and IV quadrant 2) 7 and 1st quadrant

- 3) -7 and IV quadrant 4) $\frac{1}{7}$ and 1st quadrant

42. The equation whose roots are 0, 0, 3, -4 is

- 1) $x^4 + x^3 + 12x^2 = 0$ 2) $x^4 + x^3 - 12x^2 = 0$ 3) $x^4 - x^3 + 12x^2 = 0$ 4) $x^4 - x^3 - 12x^2 = 0$

SECTION – 2 (Maximum Marks : 24)

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43. The vertices of a triangle are $(1, \sqrt{3})$, $(2 \cos \theta, 2 \sin \theta)$ and $(2 \sin \theta, -2 \cos \theta)$ whose $\theta \in R$.

The locus of centroid of the triangle and also find radius and centre

- 1) $(1, 2), 8$ 2) $(1, 2), 4$ 3) $(1, \sqrt{3})$ 4) $2\sqrt{2}, (x-1)^2 + (y-\sqrt{3})^2 = 8$
44. If 2, 3 are roots of the reciprocal equation $6x^5 - 29x^4 + 2x^3 + 2x^2 - 29x + 6 = 0$ then find other roots

- 1) ± 1 2) -1 3) $-1, \frac{1}{2}, \frac{1}{3}$ 4) NT

45. If $0 \leq x \leq \pi$, $81^{\sin^2 x} + 81^{\cos^2 x} = 30$. If $81^{\sin^2 x} = y$ then $y = \underline{\hspace{1cm}}$

- 1) 3 2) 27 3) 327 4) NT

46. Let (α, β) be the centroid of the triangle formed by the lines $15x - y = 82$, $6x - 5y = -4$ and $9x + 4y = 17$, the $\alpha + 2\beta$ and $2\alpha - \beta$ is

- 1) 6 2) 7 3) (6, 7) 4) NT

47. If $\frac{\sin^4 x}{2} + \frac{\cos^4 x}{3} = \frac{1}{5}$ then

- 1) $\tan^2 x = \frac{2}{3}$ 2) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{1}{125}$ 3) $\tan^2 x = \frac{1}{3}$ 4) $\frac{\sin^8 x}{8} + \frac{\cos^8 x}{27} = \frac{2}{125}$

48. Which of the following numbers are irrational?

- 1) $\sin 15^\circ$ 2) $\cos 15^\circ$ 3) $\sin 15 \cos 15$ 4) $\sin 15 \cos 75$

SECTION - 3 (Maximum Marks : 24)

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49. The value of $\log(\sin 1^\circ) \cdot \log(\sin 2^\circ) \dots \log(\sin 179^\circ) = \underline{\hspace{2cm}}$
50. A(a, b) and B(0, 0) are two fixed points. If m_1 is the mid point of $\overline{AB}$, m_2 is midpoint of $\overline{Am_1}$, m_3 is midpoint of Am_2 and so on then m_5 is $\underline{\hspace{2cm}}$
51. The value of b so that $x^4 - 3x^3 + 5x^2 - 33x + b$ is divisible by $x^2 - 5x + 6 = \underline{\hspace{2cm}}$
52. If (a, b, c) are in A.P. Then the straight line $ax + by + c = 0$ will always passes through fixed point whose x-coordinate is $\underline{\hspace{2cm}}$
53. Find the value of $\tan 1^\circ \tan 2^\circ \dots \tan 89^\circ = \underline{\hspace{2cm}}$
54. Number of integral points exactly in the interior of the triangle with vertices (0, 0) (0, 15) and (15, 0) is $\underline{\hspace{2cm}}$